Aahil Khan

B.E. Computer Engineering
Thapar Institute of Engineering & Technology

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Skills

Languages: Python, TypeScript, JavaScript
Backend: Node.js, FastAPI, Flask
Frontend: React, Next.js
Databases: PostgreSQL, MySQL, Redis
Cloud & DevOps: AWS (EC2, Lambda, API Gateway, DynamoDB, S3), Docker
AI/ML: LLM APIs, RAG pipelines, embeddings, semantic search

Professional Experience

Full Stack AI Intern — Oddmind Innovations *Bangalore, India (Remote)*
Next.js, React, TypeScript, Node.js, AWS, LiveKit Apr 2026 – Present

- Delivered features across frontend, backend, and real-time services for a live AI interview platform, contributing to systems that have processed **100,000+ interview minutes**.
- Developed an audio based interview evaluation pipeline using **AWS S3 and Gemini**, including multidimensional scoring and outcome generation.
- Built multi organization prompt versioning and team management workflows supporting organizations with up to **500 members**.

Team Lead / Full Stack Engineer — Thapar EduTube *Patiala, India*
Next.js, React, TypeScript, Node.js, PostgreSQL, Prisma, Redis, Docker Oct 2024 – Jan 2025

- Led a team of 3 engineers for a production lecture platform serving **21,000+ students and faculty**, owning the full stack from database design to deployment and monitoring.
- Built a **TypeScript CLI** for non-technical staff to publish lectures via synced folders, with resumable uploads and idempotent APIs.
- Simplified search using PostgreSQL full-text and fuzzy indexing; deployed with Docker, Nginx, and Prometheus/Grafana monitoring, validating **sub-800 ms p95 latency** through load testing.

Backend Engineer — Datesheet Generator, Thapar University *Patiala, India*
Python, Flask, MySQL Mar 2025 – Jul 2025

- Redesigned the system into a persistent multi-user workflow and improved scheduling, seating, and invigilator allocation with **automated validation and regression tests**.
- Built its containerized production deployment, monitoring, access controls, and Oracle PeopleSoft integration specification to automate university data workflows.

Selected Projects

SOP Opera — Agentic Industrial Safety Intelligence Platform *(Economic Times AI Hackathon 2.0 Winner)*
Python, TypeScript, FastAPI, LangGraph, PostgreSQL/pgvector, Next.js, Docker

- Built a risk engine combining sensor, permit, isolation, and workforce data; reduced false negatives from **44.5% to 0% across 593 cases** and detected hazards up to **28 minutes earlier.**
- Designed a conditional LangGraph multi-agent pipeline with hybrid pgvector/SQL retrieval and citation validation while keeping safety verdicts deterministic and LLM outputs advisory.

FlowSync — Persistent Memory for AI Coding Agents
TypeScript, Python, AWS (Bedrock, Lambda, DynamoDB, S3), MCP, Next.js

- Led backend and integration, deploying a serverless AWS stack with CDK, API Gateway, Lambda, S3, and DynamoDB.
- Built an MCP server and VS Code extension for automatic context capture, project search, and branch-aware RAG.
- Hardened the Bedrock retrieval pipeline by adding caching, model fallback, and CloudWatch telemetry, achieving **1.26s median push-to-search latency.**

Holt — Evidence Based Repository Evaluation for AI Coding Agents *(Micro1 Frontier Engineering Challenge Award Winner)*
Python, Textual, GitHub GraphQL

- Created an open source TUI that evaluates whether GitHub repositories are ready for contributors using a five stage LLM pipeline and deterministic verdict engine
- Built a contamination resistant evaluation framework using hash pinned datasets, temporal holdouts, and pre registered rules; achieved **0.63 out of sample MCC versus 0.21** for a single prompt baseline.

Konta — Context-Aware Browsing System *(Samsung PRISM Web Agent Hackathon Winner)*
TypeScript, React, IndexedDB, Transformers.js, ONNX

- Built a local first Chrome extension that converts browsing activity into an on-device knowledge base using **on-device semantic embeddings** for private, low latency search.
- Engineered **semantic/temporal knowledge graphs** and sessionization to connect related pages while preventing unrelated browsing tasks from merging.

Achievements

1st Place — **Economic Times AI Hackathon 2.0** *(60,000+ participants, 10,000+ teams, 28 finalists)*
Most Useful Real World Workflow Award — **Micro1 Frontier Engineering Challence 2026** *(8000+ teams)*
Winner — **NioHack 2026, Niograph Inc., USA** *(100+ teams)*
Winner — **Samsung PRISM Web Agent Hackathon** *(450+ participants, 120+ teams)*
Innovation Award — **Agentic AI Hackathon 2025, Ulster University, UK** *(150+ teams)*

Education

Thapar Institute of Engineering & Technology *CGPA: 9.16*
Bachelor of Engineering, Computer Engineering Aug 2023 – Present

Indian Institute of Technology Madras *CGPA: 8.34*
Diploma in Programming (concurrent online CS diploma, run alongside the B.E.) Sept 2023 – Sept 2025